

Linemod structure

Linemod_preprocessed

In total 15 objects (missing objects 2, 3, 5 and 7) and we are also in the code discarding objects 3 and 7.

- Data
 - A folder for each object
 - Each folder contains:
 - Depth images (.png) → very dark images (made code to better visualize them by normalizing to fit in 0-255 range)
 - Mask images (.png) → only one mask per image
 - RGB images (.png) → the raw image itself
 - Gt.yml → ground truths information. For each frame, we get information about:
 - object id → links frame to the corresponding 3D object model
 - object bounding box coordinates (in 2D) → → proves region of interest (ROI) in the image where the object is located
 - cam_R_m2c (3x3 Rotation matrix) → defines the orientation of the object in camera space
 - Cam_t_m2c (3x1 Translation vector) → describes the translation of the object's center from the camera to the object in millimeters
 - cam_R and cam_t → define the 6D pose of the object
 - Info.yml → contains for each frame the camera intrinsics information (K matrix) which is the same for all frames and the depth_scale
 - Test.txt → states all frames that should be used during testing
 - Train.txt → states all frames that should be used during training
- Models
 - Contains all 3D models of each object in a .ply format (including the 2,3,5,7 which were missing in the data folder)
 - Models_info.yml → metadata about the 3D models (diameter of bounding box, min_x, min_y, min_z (minimum coordinates of object's bounding box along respective axes), size_x, size_y, size_z (length, width, height of the object's bounding box along respective axes))
- Segnet_results
 - We have a folder for each of the objects which contains .png images of the object mask predicted by SegNet (not really used, but rather given to show as comparison for the masks given already in data folder)

Demo data structure

Demo_data

- kinect_driller_seq
 - Depth → images (.png)
 - Masks → only one image of the object's mask (.png). Object being in the initial position before being moved
 - NOTE: this is demo data so when we want to work with our own dataset we will need a mask for each frame if we are to use FoundationPose
 - Mesh → we have information about the 3D model itself
 - Material_0.png → texture of the object
 - Material.mtl → information about the material type
 - Textured_mesh.obj → actual 3D model
 - Ob_in_cam → contains .txt files for each frame with information about their transformation matrix (rotation matrix, translational vector, and homogeneous coordinate row)
 - In other words, this would be the gt.yml from Linemod providing the actual transformation matrix of the object
 - But this is also created in the debug folder once model is done
 - Rgb → folder containing all raw images to be fed in the model
 - Track_vis → folder which contains all the RGB images as in the rgb folder but also provided together with their 3D bounding box and x,y,z orientation
 - This is also created in the debug folder once the model is done
 - cam_K.txt → contains for each frame the camera intrinsics information (K matrix)
- Mustard0
 - Depth → images (.png)
 - Masks → only one image of the object's mask (.png). Object being in the initial position before being moved.
 - NOTE: this is demo data so when we want to work with our own dataset we will need a mask for each frame if we are to use FoundationPose
 - Mesh → we have information about the 3D model itself
 - Texture_map.png → texture of the object
 - Textured_simple.obj.mtl → information about the material type
 - Textured_simple.obj → actual 3D model
 - Rgb → folder containing all raw images to be fed in the model
 - cam_K.txt → contains for each frame the camera intrinsics information (K matrix)

Conclusion:

The necessary elements that are imperative to the model are: Depth images, Masks (or bounding boxes), Mesh, RGB, cam_K.txt